PassFrame tuition

Volume & 3D Shapes

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: What is the volume of a cuboid measuring 5cm × 3cm × 2cm?
Method & answer: 5 × 3 × 2 = 30 cm³.

Standard: A cuboid has a volume of 48cm³. Its length is 6cm and width is 4cm. What is its height?
Method & answer: 48 ÷ (6 × 4) = 48 ÷ 24 = 2 cm.

Advanced: Convert 2500cm³ to litres (1000cm³ = 1 litre).
Method & answer: 2500 ÷ 1000 = 2.5 litres.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the volume of a cuboid measuring 4cm × 3cm × 2cm?

2. What is the volume of a cube with sides of 3cm?

3. Which of these has a volume of 30cm³? A) 5×3×2 B) 6×5×1 C) both A and B D) neither

4. How many edges does a cube have?

5. How many vertices does a square-based pyramid have?

6. What is the volume of a cuboid measuring 6cm × 3cm × 2cm?

7. Which 3D shape has 1 curved face and 1 flat circular face, coming to a point? A) cylinder B) sphere C) cone D) cube

8. What is the volume of a cube with sides of 4cm?

9. Select ALL of these that are properties of a cuboid: A) 6 faces B) 12 edges C) 8 vertices D) 1 curved face E) all faces are rectangles

10. What is the volume of a cuboid measuring 8cm × 2cm × 1cm?

11. How many edges does a triangular prism have?

12. Which shape is made from 2 congruent triangular faces and 3 rectangular faces? A) cube B) triangular prism C) square-based pyramid D) cylinder

13. How many faces does a triangular prism have?

14. What is the volume of a cuboid measuring 9cm × 1cm × 2cm?

15. How many vertices does a triangular prism have?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A cuboid has a volume of 60cm³. Its length is 5cm and width is 4cm. What is its height?

2. Convert 1000cm³ to litres (1000cm³ = 1 litre).

6. Which net folds to make a cube? A) 6 squares arranged in a cross shape B) 6 triangles C) 4 rectangles and 2 triangles D) 2 circles and a rectangle

7. A fish tank measures 20cm × 10cm × 15cm. What is its volume in cm³, and in litres?

11. A cube's 6 faces are identical squares, each with an area of 16cm². What is the length of one edge?

12. Which shape has the SAME volume as a 6cm×2cm×2cm cuboid? A) a 3cm×4cm×2cm cuboid B) a 4cm×4cm×2cm cuboid C) both A

and B D) neither

3. Which of these has a volume of MORE than 40cm³? A) 5×4×2 B) 6×3×2 C) both A and B D) neither

4. A cuboid measures 7cm × 2cm × 2cm. What is its volume?

5. Convert 3000cm³ to litres.

8. Select ALL of these cuboids that have a volume between 30cm³ and 50cm³ (exclusive): A) 5×4×2 B) 6×3×2 C) 4×4×3 D) 10×2×1 E) 3×3×5

9. Cuboid A measures 4×3×2cm; cuboid B measures 3×3×3cm. Which has the greater volume?

10. A cuboid's volume is 96cm³. Its length is 8cm and height is 3cm. What is its width?

13. A cuboid has dimensions 7cm × 2cm × 3cm. What is its volume?

14. Convert 2.5 litres to cm³.

15. A cuboid measures 5cm × 5cm × 2cm. What is its volume?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A tank measures 20cm × 15cm × 10cm and is one third full of water. How many cm³ of water are in the tank?

2. A cuboid box has a volume of 120cm³, with length 10cm and width 4cm. What is its height, and how many of these boxes fit exactly into a larger box of volume 480cm³?

3. A cube-shaped box has a volume of 216cm³. What is the length of one edge, and what is the box's total surface area?

4. A pool is 8m long, 5m wide and 1m deep throughout. How many cubic metres of water fill it, and how many litres is this (1m³ = 1000 litres)?

5. A cuboid has a square base with side 4cm and a volume of 80cm³. What is its height?

6. Select ALL of these statements that are TRUE about a 6cm×4cm×3cm cuboid: A) its volume is 72cm³ B) it has 8 vertices C) its volume is 13cm³ D) it has 6 faces E) it has a curved surface

7. A tank measures 30cm × 20cm × 15cm and is filled to a depth of 10cm. How many cm³ of water does it contain?

8. A cuboid's length, width and height are in the ratio 2:2:1. If the height is 3cm, what is the volume?

9. A cube has a volume of 125cm³. It is melted down and recast into a cuboid with a base of 5cm × 5cm. What is the height of the new cuboid?

10. A tank measures 1m × 1m × 0.5m and is filled at 100 litres per minute. How long does it take to fill completely? (1m³ = 1000 litres)

11. A 15cm×10cm×5cm box is packed with 5cm cubes, with no gaps or overhang. How many small cubes fit inside?

12. A cuboid measures 8cm × 6cm × 4cm. It is cut exactly in half, dividing its 4cm height into two equal layers. What are the dimensions of each half?

13. A cuboid has a volume of 200cm³. Its length is twice its width, and its height is 4cm. What are its length and width?

14. A 15cm×12cm×10cm container, completely full of water, is poured into empty cube-shaped bottles of side 5cm. How many bottles can be completely filled, and how much water is left over?

15. A cuboid box has three different face areas: 15cm², 12cm² and 20cm². What is the volume of the box?